

# Lessons Learned from Experimental Prototype Development for KPI Acquisition and Modeling from Legacy ERP Systems

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**Abstract.** In this paper we consider a business case of an Enterprise Resource Planning (ERP) system vendor who needs to introduce qualitative features without making great investments in the legacy source code. We propose to extend legacy ERP system's functionality by applying the Leading System paradigm so that business process management can be introduced. This enables business process awareness and provides opportunity for qualitative feature set introduction in a legacy ERP system. As a result, Key Performance Indicator (KPI) acquisition, modeling and analysis from the process aware legacy ERP system becomes possible. The main focus in this paper is on lessons learned about a suitable prototype architecture for KPI acquisition, modeling and analysis, and about best practices for KPI analysis from log data of business process execution.

**Keywords:** Legacy ERP systems · Process awareness · Key performance indicators modeling

## 1 Introduction

In this paper we consider the research problem that was brought by the Enterprise Resource Planning (ERP) system vendor who wanted to introduce qualitatively new features in the ERP system without making great investments in the legacy source code. This is a rather common situation – over time ERP systems become more and more complex and at some point the vendor has to decide whether the system is to be substantially re-architected or external components are to be introduced to enable more flexible and extensible solutions. The lack of flexibility and extensibility may hinder competitiveness of client enterprises and they may have to decide whether to stay with the current ERP system vendor or to make, usually substantial, financial investments in evaluating offers of other vendors and migrating to a new system. Gradual evolution or continuous improvement have been advocated as more business friendly strategies to compare with re-engineering approaches [1]. Therefore, the motivation for this research is to provide options how to extend the capabilities of legacy ERP system with advanced features.

In particular, we focus on how to introduce Key Performance Indicator (KPI) acquisition, modeling and analysis in the context of legacy ERP systems.

Let us illustrate the general features of a business case which we use to illustrate the problem. An ERP system vendor has many customers, mostly, small and medium size enterprises. Each customer uses the ERP solution for its specific needs, integrates it with its own and/or third party information systems (IS) and other ERP systems. It is quite common for a business process (BP) to span among several participants from different enterprises, which therefore must collaborate to obtain business value by using multiple ISs. The particular ERP vendor, however, has limited possibility to change or enhance the functionality of the third party ISs. Also in the context of legacy ERP systems, workflows supported by process engines might not be directly applicable. Thus, when the knowledge about business processes is only in the heads of end users or at the best in some decoupled descriptions, understanding and managing how process instances are being executed in reality becomes the hardly achievable task. Therefore, we assume that if it would be possible to detect changes in the ERP system's business objects externally, they could be used to provide user guidance and also to perform monitoring of KPIs from the log data of executing business processes.

The research objective is to provide an approach for extending the capability of legacy ERP systems with KPI acquisition, modeling and analysis functionality. The constraint to be taken into consideration is that in legacy ERP systems there is no clear mechanism how to obtain KPIs from business process execution log data; and although experienced users might know their tasks and execution order, they still are not able to analyze current and past business process execution characteristics.

As a research method we use prototyping and lessons learned format of presentation by grouping research questions into categories and for each research question stating an issue, analyzing a problem, and providing summarizing recommendations. From the research objective we have identified eight research questions and for each give a literature review, applied approach, motivation and achieved results.

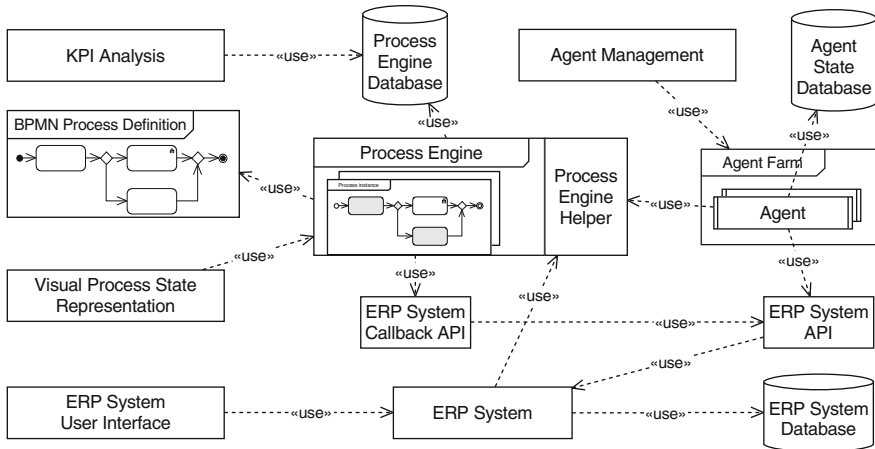
The rest of the paper is organized as follows. Section 2 outlines general architecture of the prototype with which we have modeled KPI data. Section 3 provides detailed discussion about the role of KPIs. Section 4 discusses KPI acquisition methods from business process simulation tools. Section 5 discusses KPI analysis and visualization tools and demonstrates example obtained from the implemented prototype. Finally, Sect. 6 concludes the paper with the summarization of main results and outlines future work.

## 2 General Architecture of the Prototype

In this section we provide assumptions and detailed description regarding the general architecture of the prototype that was used during the modeling and experimental development for obtaining KPIs.

**Research Question 1: Is it possible to validate specific components of the prototype against the general architecture in relation to KPI acquisition, modeling and analysis?**

*Procedure.* We assume that the proposed prototype must be based on an architecture as described in [2]. The authors of [2] have evaluated different business modeling



**Fig. 1.** General architecture of the prototype for KPI acquisition and modeling.

paradigms and concluded that for the non-disruptive introduction of process awareness (which is essential for KPI acquisition, modeling and analysis) the Leading System paradigm is the most optimal conceptual tool, though it comes with limitations on the business process definitions. The general architecture of the prototype is depicted in Fig. 1.

The following are the main components of the architecture:

- **Process Engine** – the main task of which in our context is to react on signal messages sent by external agents, call Service Tasks and maintain process definitions and their states. We use Camunda<sup>1</sup> process engine.
- **(Legacy) ERP System** – ensures business data processing via user interface and external Application Programming Interface (API) for agents and the process engine in order to obtain and modify the database of ERP system.
- **Agent Farm** – by accessing the API of ERP System, agents monitor the business data for specific states and notify process engine when predefined conditions are met.

Initially, the process engine was embedded in the web application that ran in the context of JavaEE application server. This simplified prototype implementation and testing. For example, by default the embedded process engine used H2 in-memory database. Therefore it was possible to start with new process engine structure each time it was launched. In order to test in a production-like environment, however, the process engine was implemented as a separate application that uses persistent storage and provides callback API to the ERP system. The goal of the callback API is to provide Business Process Model and Notation (BPMN) Service Tasks with the ability to call ERP system API. During the process execution the process engine

<sup>1</sup> <http://camunda.org>. The particular selection is not in the scope of this paper, however it should be noted that there was done evaluation among multiple process engines.

creates an instance of ERP system callback API component, makes calls to its operations and ensures that changes are committed in the ERP system database.

To provide process instance advancements and state changes, we assume that initiators for new process instance creation and existing instance advancement are end users of the ERP system. For example, let's consider a case when user registers new invoice via the ERP system's user interface. This ultimately leads to changes in the ERP system's database (for example, new records in the tables are created). As a result of these activities, an agent detects these changes and initiates sending a signal message to the Process Engine Helper (PEH) component. The PEH component ensures that in the process engine either new process instance creation or existing instance advancement is performed. To ensure this behavior, there must be relationship between the ERP system's data objects and the business process that correlates to these data objects. Agents and the PEH component then only have to ensure proper synchronization between the ERP system's database and the process engine database. We call it the Leading System paradigm and it allows us to introduce process awareness for legacy ERP systems [2].

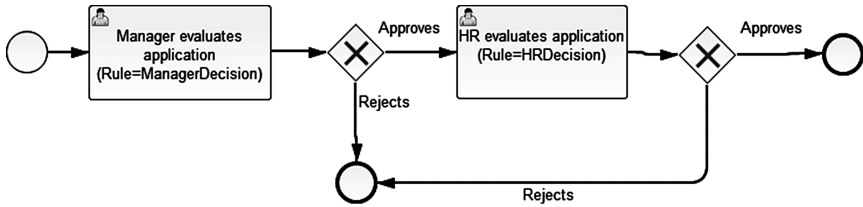
From the process engine viewpoint, the behavior of the process instance must map to the data model in such a way that, upon receiving signal messages from the ERP system, the advancement of the process instance is made (it is ensured by external agents and the PEH). This behavior can be modeled with such elements of BPMN 2.0 standard as User Task, Receive Task, Timer and Catch Events. For example, when the process instance state reaches a particular User Task element, the process engine creates a new task instance which must be completed for the process instance to advance further. Such task can be related to a particular user or role in the ERP system. Receiving a signal message indicates that a particular task is completed, the process instance is advanced accordingly (i.e., in the ERP system database the state has changed and therefore detected by an agent rule). It is possible not only to receive the signal message about the fact that a particular task has been completed, but also to attach related data to it. For example, this data can be mapped to the process instance variable. Therefore it is possible to make branching decisions during a process instance execution (modeled as BPMN XOR Gateway elements).

To simplify synchronization between the ERP system and the process engine, we enforce the constraint that each process definition must be related to only one business entity (object). With a business entity we denote anything that has a unique business key (identifier) and during its lifecycle can be in different states. As a result, there can be multiple business process instances, but each is related to the unique business entity. For example, business entity "Invoice" can be related to and used in the "Invoice management process."

Another important aspect is that the PEH is able to return the state of successfully applied signals and also to show, which signals were not applied. This information is used by agents in order to not resend the same state signal messages multiple times.

Let's review how these principles can be applied for modeling a simple coordination business process as shown in Fig. 2. The testing of this model can be performed for the following execution scenario:

1. The PEH receives a signal message (containing a business entity name and an identifier) that corresponds to the start event rule of the process definition. As a result,



**Fig. 2.** An example of a simple coordination business process

the new process instance is created and automatically advanced to the first user task – “Manager evaluates application.”

2. Next, the PEH receives a signal message from an agent that the rule “ManagerDecision” has been completed. This message contains also data about the decision (approved or rejected).
3. Assuming that the approval decision was made, the process instance advances and stops at the second user task “HR evaluates application.”
4. Next, the PEH receives another signal message from an agent that the rule “HRDecision” has been completed. This message contains also data about the decision.
5. Assuming that also the second approval decision was made, the process instance advances to the end state.

To provide support for BPMN Service Tasks we assume that process definitions can contain references to Java classes, which during process execution are instantiated, and via the callback API perform automated activities in the ERP system.

However, the important aspect of the process engine behavior is that it saves details about each process instance execution history in its database. This information then can be used as a base for KPI acquisition, modeling and analysis. This aspect is the main subject of discussion in further sections.

*Summary.* By applying the Leading System paradigm, the prototype application was developed and placed between the ERP system and the process engine. We created various BPMN process definitions and by applying process testing classes, we performed unit and integration tests. Unit and integration tests mostly simulated agents that send signal messages to the process engine thus creating/advancing/ending process instances and providing the feedback to the ERP system.

To monitor process engine states we used Camunda Cockpit web application. It allows visualization of information about executing process instances and their states. This information is important not only for evaluation of the current status of the process engine from different perspectives, but also KPI and Business Intelligence (BI) analysis can be performed on the base of this information.

### 3 Key Performance Indicators

In this section we give general definitions of Key Performance Indicators (KPI) and their usage in business process analysis and modeling.

**Research Question 2: What is KPI and how to use them in BP performance monitoring?** The developed prototype foresees business performance monitoring. Therefore one of the tasks is to review the principles of KPIs and their usage in business process analysis and modeling.

*Procedure.* We have studied KPI literature and their use in business process monitoring solutions. KPIs are quantifiable measurements, agreed beforehand, that reflect the critical success factors of an organization [3]. In other words, they are a set of quantifiable measures that a company or industry uses to gauge or compare performance in terms of meeting their strategic and operational goals [4]. KPIs are used in areas like revenue improvement, cost reduction etc. In this case, the goal is to collect business process performance KPI metrics for further business process performance modeling and analysis.

There are many of KPIs available to help to measure BP performance. Examples of such KPIs are [5]:

- Percentage of processes where completion falls
- Average process overdue time
- Percentage of overdue processes
- Average process age
- Percentage of processes where the actual number of assigned resources is smaller than planned number of assigned resources
- Sum of costs of stopped active processes
- Average time to complete a task
- Sum of deviation of time (for example, in days) against the planned schedule of all active projects

Different opinions exist about the types and quantity KPIs needed for successful BP performance measurement [6]. For example, Hope and Fraser [7] recommend to use no more than ten KPIs. Others, like Kaplan and Norton [8] recommend using no more than twenty KPIs. We see that in practice, the amount and specific features of KPIs depend on the specific features of the organization and goals. Another way is to use the rule “10/80/10” [9]. It means that an organization should have about ten Key Result Indicators (KRIs), up to eighty Performance Indicators (PIs) and Result Indicators (RIs), and ten KPIs. Parmenter [10] recommends to use one or two daily or in real-time (24/7) updated KPIs and five important KPIs that are updated once a week. In those seven KPIs, a number of KRIs are required.

KPI measures have at least four aspects: budget, scope, time and quality. Parmenter [10] indicates that a correct performance KPI must be related to one or more organizational success factors and quality models. Less important KPIs can be related only to success factors of an organization.

KPIs are used for different purposes, for example, for organization’s strategic, tactical and operational measurement management [6, 11]. It is more difficult to create KPIs for strategic levels, for example, to measure organizations strategic objectives. It is because usually strategic objectives are related to more than one business process and it means that processes differ each from other. One more difficulty is that in practice the

validation usually can be performed only when KPIs are already integrated into systems and used organizations. Only then it is possible to see whether these KPIs really give needed information. There are situations when KPIs are to be modified or changed, for example, when the organization changes its objectives or existing KPIs fail to bring relevant information for decision makers [10].

Modeling of KPIs can be carried out by at least two methods – a protocol modeling method and an alternative “attributes” method. The protocol modeling method is based on quality requirements of the organization combined with business strategic, tactical and operational goals. And the “attributes” method is based on concept of meta-type attributes like measurements. An example of such attributes might be the “average number of employees” of an organization.

*Summary.* In each situation it is recommended to use a collection of ten KPIs that show the most important measurements. Of course, there can be more than ten indicators that will help to define the most important KPIs. KPIs development can be made mathematically, but the real validation can be performed only during KPIs usage on a daily basis.

Measurements can have at least four aspects: budget, scope, time and quality. The prototype, discussed in this paper, does not cover the financial aspect, so only three other aspects will further be discussed.

In addition, KPIs have an operational and tactical perspective. In the prototype the main focus is on internal processes, specifically – on business process management and performance quality control. The prototype does not restrict the types of KPIs, for example, whether they are related to strategic, tactical or operational goals. This helps to create a solution for wide range of business processes. The prototype permits that KPI measurements can be taken on a past, present and future process executions, and it can be done regularly with different time intervals or upon request.

In the next section the tools that support KPIs creation and necessary data import are described.

## 4 Acquiring KPIs from Business Process Simulation Tools

### **Research Question 3: What best practices exist for performance metrics of BPs?**

The prototype concept foresees such business performance metrics that touch the scope, time and quality in operational and partially tactical perspectives. One of the tasks is to discover what best practices exist for business performance metrics, since they are the foundation for KPIs.

*Procedure.* We have checked specifications of four business process modeling tools, which have well-developed mechanisms for measuring business processes either during simulation or in runtime mode. These tools are Grade-BM v4.1, IBM WebSphere Business Modeler Advanced v7.0, MicroStrategy Analytics Enterprise v9.3 and v9.5, and ARIS Process Performance Manager.

Specifications of MicroStrategy [12] and ARIS [13] lack the description of available performance metrics and KPIs in details. However, ARIS suggests functionality for analysis of process instances using simulation, KPIs evaluation in online mode, as well

as data mining. KPIs touch process time, costs, resources, quality and performers. MicroStrategy defines the metric as one or more aggregate functions applied to table columns (or expressions/functions of them) at a particular level of metric [14]. MicroStrategy does not contain a predefined list of KPIs, but has two pre-packed applications that include two business-specific modules, namely, “Customer Analysis” and “Financial Analysis.” Both of them were not appropriate due to lack of required data in the prototype. Nevertheless, MicroStrategy suggests more than 300 embedded mathematical, OLAP, financial and statistical functions to help in creating KPIs at any level of complexity.

Grade-BP [15] is intended for business process modeling and simulation in order to discover bottlenecks in the processes during the design. Additionally, it is possible to ground user-specified business metrics on defined metrics in the tool and to simulate their achievement at defined resources and time periods. The accumulated statistics then can be used for analysis of tasks, events and performers. From the Grade-BM statistics view, a transaction is also a task. If applicable, Grade-BP provides aggregated measurements for one simulation session (complete or partial) in the form of total, maximal, minimal and average values in the time period.

IBM WebSphere Business Modeler Advanced v7.0 [16] foresees monitoring global processes without any code generation or deployment, at any level of detail and with or without event distribution. Processes are analyzed dynamically at runtime and during event origin. It is possible to analyze time, costs, resource allocation, “free float” activity time, the most critical and the shortest paths of processes, as well as a process case.

*Summary.* The analysis gave us the general view on the process business performance metrics provided by the MicroStrategy, Grade-BM and IBM WebSphere Business Modeler Advanced. The levels of measuring can be the following: a task, an activity instance, a set of activity instances (or an activity), a process instance, a set of process instances (or a process), a process case instance, a set of process case instances (or a process case), an event queue, a resource, and a performer. Fifty one candidate for process performance metrics have been elicited and then analyzed in more detail.

**Research Question 4: What raw data are required to collect the appropriate process performance metrics from the elicited ones?** As previously mentioned, the prototype has limitations on costs (related to financial KPIs at the enterprise units level), resources and performers characteristics. Therefore, not all of the elicited metrics may be appropriate from the available raw data. However, some of the required data potentially may be added to the process definition scheme as an extension part.

*Procedure.* During the research, raw data that may be required for each of fifty one elicited metric evaluation are identified. Those metrics, which require data that potentially are hard to be obtained from the prototype at the present, were set as inappropriate and excluded from the list.

*Summary.* The refined performance metrics include measurements at the action (a task, an activity, a transaction, a process, etc.), queue and performer levels. They include all



possible execution and delay durations, successful completeness and failure rates, deviations from the planned indicators, performers load, etc.

Based on the results of the assessment of the required raw data, we conclude that for performance metrics the prototype should be able to collect the following data:

- For activity instances and process instances: start time, end time, failed end label (an indicator that an instance finished its execution unsuccessfully), planned execution duration, and work start time
- For queues: since the prototype events also are related only to the process instances and activity instances, these metrics use instances data
- For performers: instance performer and previously mentioned data of instances.

**Research Question 5: What best practices exist for KPI templates?** The task is aimed to analyze those performance metrics which could be considered as KPIs.

*Procedure.* We have analyzed best practices in offers of KPI libraries. Since KPIs are very business function/area specific, we have searched KPIs libraries in scientific literature [17–19]; stakeholders’ communities KPI Institute [20], KPI Library [21], and APQCs Process Classification Framework [22]; and BI tools [16, 23]. A number of KPIs is huge. Therefore, we have focused only on those, which may touch quality of business process execution.

KPI Institute and KPI Library have many suggestions for KPI identification, however most of them are particular business specific. We have elicited 2 KPIs from KPI Institute and 20 KPIs from KPI Library (mostly related to project/program management).

APQCs Process Classification Framework (PCF) used in IBM WebSphere Business Modeler Advanced v7.0, is applied from enterprise levels 2.0 “Develop and Manage Product/Services” up to 8.0 “Manage Financial Resources.” Most of the presented KPIs had a close relationship with financial indicators, managerial decisions, sales/services specifics, staff costs, management of quality assurance activities, and narrow business specifics. However, as for the prototype general performance indicators were needed, we have elicited only twelve KPI candidates from more than 200 indicators. In turn, Klipfolio tool provides KPI examples for different functional areas: sales, marketing, supply chain, call centers, help desk, assurance, small enterprise, health care, social media and SEO KPIs. Unfortunately, all the provided KPIs are particular business specific. Therefore, they were not included in the KPI candidate list.

*Summary.* We have elicited 2 general KPIs from enterprise knowledge management metrics [19], 28 KPIs related to client orientation and internal processes of Balanced Scorecard areas [17], and 12 KPIs from the recommended ones in [18] at the process and sub-process levels used to measure execution bottlenecks. The result is the list of 76 KPIs candidates.

**Research Question 6: What raw data are required to collect the appropriate KPIs from the elicited ones?** The same as in case of performance metrics we required additional analysis of raw data necessary to measure KPIs.

**Table 1.** Summary of available raw data for KPI calculation from the prototype

| No. | KPI                                                                  | Data availability                                                                                    |
|-----|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 1   | Daily log                                                            | Full                                                                                                 |
| 2   | Completeness status (%)                                              | Not enough, because of missing source data and computation of process execution scenarios            |
| 3   | Work schedule achievement in the planned period                      | Partly, because can be applied only to finished processes                                            |
| 4   | Incomplete process task rate                                         | Full                                                                                                 |
| 5   | Yield (%)                                                            | Full                                                                                                 |
| 6   | Total turnaround time (for a business process)                       | Full for finished processes                                                                          |
| 7   | Total lead time                                                      | Full for finished processes                                                                          |
| 8   | Deviation from planned work hours (%)                                | Full for finished processes                                                                          |
| 9   | Deviation from the schedule (hours)                                  | Full for finished processes                                                                          |
| 10  | Delayed process activities (%)                                       | Partly, because can be applied only to finished processes                                            |
| 11  | Milestones missed in the time period (% , #)                         | Not enough, because of missing source data                                                           |
| 12  | Average delay duration of process activities                         | Not enough, because of missing source data                                                           |
| 13  | Average close duration of process activities                         | Full for finished processes                                                                          |
| 14  | Close duration rate of process activities                            | Full for finished processes                                                                          |
| 15  | Queue rate of process activities                                     | Full                                                                                                 |
| 16  | Planned hours for activities (#)                                     | Full                                                                                                 |
| 17  | Forecast accuracy one period before the production                   | Full for finished processes                                                                          |
| 18  | Incomplete work messages as well as their statuses and sizes         | Full, if process owner is the same as executor                                                       |
| 19  | Work success and failure rate                                        | Full                                                                                                 |
| 20  | A number of new process instances (#)                                | Partly, because can be applied to started processes, but can't be applied to planned start date/time |
| 21  | On time completed process (%)                                        | Full, if planned end date can be calculated dynamically                                              |
| 22  | Date when ancillary work must be completed (for main processes only) | Full for delegated activities                                                                        |
| 23  | Processes completed on time and on budget                            | No enough, because of missing data about expenses and budget                                         |
| 24  | Main work that subcontractors perform                                | Partly                                                                                               |
| 25  | Execution/performance time for top ten projects/processes            | Full for finished processes and if process executor is also process owner                            |

*Procedure.* The 76 KPIs candidates selected in the previous step have been refined with their assessment formulas and required raw data. Those candidates that required additional data on finance, costs and staff work time periods have been excluded from the final KPI candidate list. The final list contains 25 KPIs and availability of the necessary data for which are summarized in Table 1.

The required raw data the prototype should be able to collect are the following:

- For activity instances: end time, work start time, performer, category, name, number, planned end time, planned execution duration, planned start time, start time;

- For process instance: path, alternate path, end time, budget, owner, costs, performer, planned end time, planned execution time, planned start time, start time;
- For milestone instances: occur time, planned occur time.

Unfortunately not all required data could be obtained from the process engine database used in the prototype. Such data as “planned duration” of the activity and the process, and the “process owner” have to be obtained from the process definitions themselves. But data about startup time of a task, planned startup time of an activity, planned occur time for a milestone, alternative path for a process, process budget, process execution path, and process expenses are not retrievable at all.

*Summary.* We have elicited an initial list of process performance metrics and KPIs that are general enough. Both metrics and KPIs require a small set of raw data related to activities, processes, milestones, and performers/owners. All raw data can be exported by the prototype either from the runtime/historical raw data, or from the process declaration schemas. Not all KPIs are measurable within the prototype however – 17 KPIs can be measured completely, 4 only partially and for 4 KPIs the process engine does not provide enough data.

## 5 Data Import into KPI Analysis and Visualization Tools

**Research Question 7: What possibilities give tools which support KPIs analysis and visualization?** Today in the market many KPIs analysis and visualization tools that allow to operate with data from external data sources (for example other information systems) are available.

*Procedure.* We have checked specification of KPIs tools, which have support for data analysis and visualization. These tools are represented in Table 2.

Not all of these tools support OLAP, Extract, Transforms and Load (ETL), Data Mining and KPIs at the same time [24]. For example, only the following open source tools support above-mentioned possibilities – SpagoBI, Pentaho and Vanilla. Such open source tools like OpenI, JasperSoft and Palo do not support KPIs.

According to [25], the ten most recommended tools for BI are Pentaho, Sisence, Bayound Intelligence, Birst, TARGIT Decision Suite, iDashBoards, TIBCO Spotfire, IBM Cognos Business Intelligence, Quadbase Systems and Tableau.

The most widely used BI tools research [26] mentions tools like Adaptive Insights, Birst, GoodData, Jedox, Logi Analytics, Pentaho, Tableau, TIBCO, Actuate, Information Builder, Qlik, Dimensional Insight, Dundas, Jinfonet, Klipfolio, Phocas, Targit and the tools from IBM, Microsoft and SAP. The authors of this research mention that small and medium organizations mostly use dashboards and service solutions.

Performance measurements are based on data or facts stored in databases. Even in small and medium organizations there is more than one data source, so there could be need for data extraction. Therefore it is recommended to use various ETL technologies for necessary data extraction and transformation for further use in OLAP cubes and analysis.

*Summary.* The most widely supported data sources are relational databases (creating a direct connection to one or more databases), connections with web service or systems data through API (including ERP systems databases). Second most popular data imports are from MS Excel spreadsheets and flat files. Less popular is data import from XML files and connection to non-relational databases, for example, multi-dimensional databases and document databases.

Many of BI tools support ETL technology, but not always this means that they also support OLAP cubes.

**Research Question 8: Is it possible to export raw data from the prototype and to import them for KPI creation?** The task was to import data in one of the most supported format, CSV, and to create a KPI in some popular tool.

*Procedure.* Among a number of tools that support building and tracking the KPIs, MS Excel is one of the most popular tool. It allows to import data from different data sources, supports data charts, PivotTables, many statistical and mathematical functions as well as connections from one MS Excel file to other [27]. MS Excel has also risks and disadvantages related to unpredictable processing of large amount of data, versioning issues, unstable dashboards due to changes in charts, graphics or data structures, as well as uncontrolled editing that can lead to data integrity loss. However, its numerous functions, data analytics, calculations and visualization possibilities make this tool suitable for our experiments.

The procedure of using MS Excel as a KPI monitoring includes such actions as [28]: select necessary data, for example, import from the database; identify data pools for measurements; select and create graphs and charts needed to show on a dashboard; and create a spreadsheet where to place all the graphs and charts and put some filters on them, for example, pivot table charts, colored schemes/indicators, etc.

During our experiments, we selected raw data of the process shown in Fig. 2 as a plain denormalized table, then imported this data to MS Excel; selected the data we need to monitor and create PivotTable; and created filters and a chart for process activities durations.

*Summary.* Multiple experiments have been successfully completed. All 17 KPI's with the complete data set and 4 KPI's with the partially available data set have been modeled successfully. To get an insight into obtained results, an example pivot table for coordination business process KPI "Average close duration of process activities" is shown in Fig. 3.

Table 2. Summary of Business Intelligence tool comparison

| BI tools                         | Technologies          |                                |                    |                       |                      |     |      |     |
|----------------------------------|-----------------------|--------------------------------|--------------------|-----------------------|----------------------|-----|------|-----|
|                                  | Connection with RDBMS | Connection with DBMS dimension | Excel spreadsheets | Flat file (text, CSV) | Web Services and API | XML | JSON | ETL |
| Actuate BIRT                     | V                     | V                              | V                  | V                     | V                    | V   |      |     |
| Adaptive Insights                | V                     |                                | V                  |                       | V                    |     |      | V   |
| Beyound Intelligence             | V                     |                                | V                  |                       | V                    | V   |      | V   |
| Birst                            | V                     | V                              |                    | V                     | V                    | V   | V    | V   |
| iDashBoards                      | V                     |                                | V                  |                       | V                    | V   | V    | V   |
| Dimensional Insight The Driver   | V                     |                                | V                  | V                     | V                    | V   |      | V   |
| Dundas                           | V                     | V                              | V                  | V                     | V                    | V   |      | V   |
| GoodData                         | V                     | V                              |                    | V                     | V                    | V   | V    |     |
| IBM Cognos Business Intelligence | V                     | V                              | V                  |                       | V                    |     |      | V   |
| Information Builder              | V                     | V                              |                    | V                     | V                    | V   | V    | V   |
| Jedox                            | V                     |                                | V                  | V                     | V                    | V   |      | V   |
| Klipfolio                        | V                     |                                | V                  | V                     | V                    | V   | V    |     |
| MicroStrategy                    | V                     |                                | V                  | V                     | V                    |     |      | V   |
| Pentaho                          | V                     | V                              | V                  |                       |                      | V   |      | V   |
| Phocas                           | V                     |                                | V                  | V                     | V                    |     |      | V   |
| QlikView                         | V                     |                                | V                  | V                     | V                    | V   |      | V   |
| Quadbase Systems                 | V                     |                                |                    |                       | V                    |     |      |     |
| Sisence                          | V                     |                                | V                  | V                     | V                    | V   |      | V   |
| SpagoBI                          | V                     | V                              |                    |                       | V                    |     |      | V   |
| Tableau                          | V                     | V                              | V                  | V                     | V                    |     |      | V   |
| TARGIT                           | V                     | V                              | V                  | V                     | V                    |     |      | V   |
| TIBCO                            | V                     | V                              | V                  | V                     |                      |     |      | V   |
| Vanilla                          | V                     | V                              |                    |                       | V                    |     |      | V   |

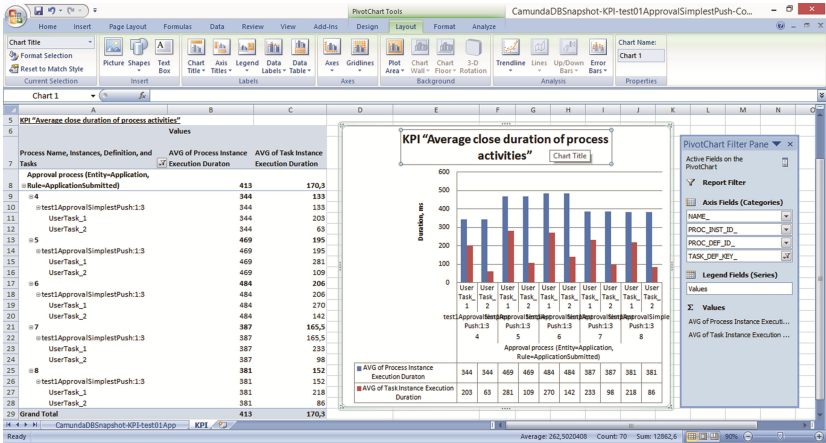


Fig. 3. The resulting pivot table for KPI “Average close duration of process activities”

## 6 Conclusions

In this paper we described our experiences and lessons learned from the experimental prototype development for KPI acquisition, modeling and analysis in the context of legacy ERP systems for which the process awareness has to be introduced.

It was shown that by implementing the Leading System paradigm, it is possible to introduce BP execution support for such legacy ERP systems that originally do not have explicit process awareness thus providing new qualitative features without making huge investments in the legacy source code. The implemented prototype allowed us to acquire additional data about business process executions and use them for KPI modeling and analysis.

One of the research tasks was to discover what kind of KPI metrics exist and what are the best practices for BP performance modeling. There are many opinions regarding types and quantity of KPIs that are needed for successful BP performance measurement and how to correctly apply them. However, usually it is recommended to use a selection of no more than ten KPIs that provide the most important measurements.

We have elicited original list of 76 KPIs from related works. This list then have been refined with their assessment formulas and required/available raw data. KPI candidates that required additional data on finance, costs and staff work time periods were excluded. Thus the final list for modeling and analysis consisted of 25 KPIs. Unfortunately not all required data could be obtained from the process engine database to calculate these KPIs: 17 KPIs can be measured completely, 4 only partially and for 4 KPIs the process engine does not provide enough data. The 21 KPI with the complete and partial data sets has been modeled successfully.

Among the tools that support building and tracking the KPIs we have selected MS Excel that was the simplest tool fitting our requirements for data input and representation. During the experiments, we selected raw simulation data of the multiple BP executions, imported these data to the Excel spreadsheet and created calculations, filters and pivot charts from process activities.

The results obtained show that for the legacy ERP system vendor it is possible to introduce additional services substantially improving ERP system's feature set that can be used both for existing and new customers. We hope that our conclusions will be helpful also for the enterprise modeling and simulation community. The future work is related to the demonstration of results to business and system analysts and based on their feedback elaboration of specific KPIs into more dynamic visualization environments, like dashboards.

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